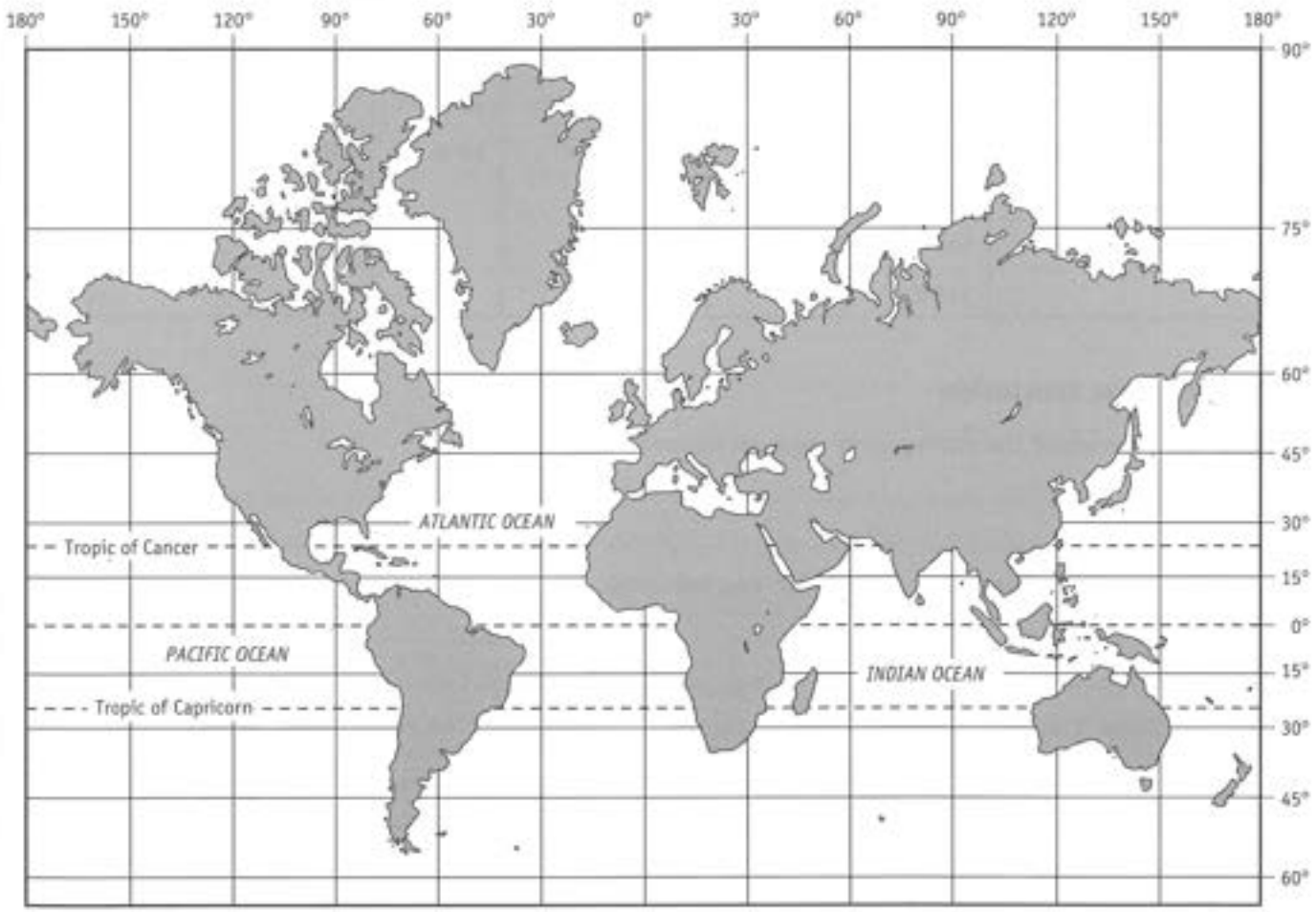
Why are earthquakes and volcanoes where they are? 20a

Labelling a map of the earth

1 Label the map below using the underlined names and coordinates from the following information.

- The equator lies between the tropic of Cancer and the tropic of Capricorn. Its latitude is 0°.
- Starting at the equator the latitudes increase by 15°N in a northerly direction and by 15°S in a southerly direction.
- The latitude immediately above the tropic of Cancer is 30°N, the latitude immediately below the tropic of Capricorn 30°S. The South Pole has a latitude of 90°S.
- The longitude passing through London is called 0°.
- Beginning at 0° the latitudes increase by 30°E towards the east and by 30°W towards the west.
- Sydney in Australia has a latitude of 150°E. Alexandria in Egypt has a latitude of 30°E and a longitude of 30°N.
- The main landmasses are Africa, Asia, Australia, Europe, North America and South America.
- Label all other longitudes and latitudes.



Note: The southern and northern ends of the earth have been stretched sideways so that a round earth fits onto a rectangular map. As a result the northern island Greenland appears much bigger on the map than it really is.

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Why are earthquakes and volcanoes where they are? 20b

Marking places of earthquakes

2 Place a red dot (•) on the map for the following major earthquakes.

Longitude	Latitude	Longitude	Latitude
20°W	70°N	80°E	10°S
40°W	40°N	20°E	40°N
70°W	60°S	60°E	20°S
75°W	30°S	70°E	30°N
75°W	40°S	80°E	10°S
75°W	50°S	90°E	20°N
90°W	0°	100°E	0°
110°W	20°N	110°E	30°N
115°W	30°N	130°E	40°N
120°W	50°N	140°E	0°
125°W	20°N	140°E	10°S
150°W	60°N	140°E	20°N
		160°E	50°N

Marking places of volcanic eruptions

3 Place a blue cross (x) on the map for the following major volcanic eruptions.

Longitude	Latitude	Longitude	Latitude
20°W	70°N	40°E	40°N
30°W	15°N	50°E	15°N
30°W	60°S	60°E	10°S
40°W	50°N	60°E	20°S
75°W	25°S	80°E	10°S
80°W	0°	100°E	10°N
80°W	10°N	130°E	30°N
100°W	25°N	140°E	10°S
115°W	30°N	140°E	40°N
150°W	60°N	140°E	45°N
		160°E	50°N

Making a first conclusion

4 a State where the earthquakes are in relation to the volcanoes.

b Describe where most earthquakes and volcanoes are.

c Suggest an explanation why the earthquakes and volcanoes are where they are.

Marking mountains and valleys

5 On the map above, draw thick lines in pencil to indicate where the following mountains and valleys are.

- The Rockies are along the west coast of North America.
- The Andes are all along the west coast of South America.
- The Himalayas are north of India on 30°N from 70°E to 90°E.
- The Urals separate Asia and Europe and run from 50°N to 70°N along 60°E.
- The Mid-Atlantic Ridge, which is continuously being formed, is on 20°W from 60°S to 90°N.

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Why are earthquakes and volcanoes where they are? 20c

Drawing a second conclusion

6 a Describe where the mountains and valleys are compared to the volcanoes and earthquakes.

_____

_____

b Explain why the mountains and valleys are where they are. (Hint: refer to information on plate tectonics.)

_____

_____

c Explain what the tectonic plates are doing near the west coasts of North and South America.

_____

_____

d Explain what the tectonic plates are doing at the Mid-Atlantic Ridge.

_____

_____

7 Give the coordinates of the following locations:

a Cuba _____

b Iceland _____

c India _____

d Italy _____

e Japan _____

f Korea _____

g Madagascar _____

h New Zealand _____

i Papua New Guinea _____

j Spain _____

Extension exercise – Adding tectonic plates to the map

8 Refer to worksheet 18 *Plate Tectonics* to answer the following questions.

a In blue sketch the borders of the main tectonic plates on the map from question 1.

b Label the main tectonic plates.

c Indicate the subduction zones by drawing a red line along the relevant plate boundaries.

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